

Position-Resolved Likelihood Inference for Atom Gradiometer Image Data

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1 Setup and Notation

1.1 Gradiometer Architecture

We consider an atom gradiometer composed of two atom interferometers (AIs) operated simultaneously, one with atoms sourced at height $z = 0$ m and one at $z = 100$ m, indexed hereafter by $z \in \mathcal{Z} = \{0, 100\}$. Each AI independently measures the accumulated phase along its atom trajectories. Common-mode laser-phase noise cancels in the differential signal, while a spatially-dependent differential phase (e.g. from a gravitational wave or dark-matter field) is retained.

A run consists of N_{shots} shots indexed $i = 1, \dots, N_{\text{shots}}$, acquired at times t_i . Each shot produces two 2D fluorescence images per AI: one for ground-state atoms ($s = g$, bright port) and one for excited-state atoms ($s = e$, dark port). Images share a common pixel grid with bin centres $\mathbf{x}_b = (x_b, y_b)$, bin area $\Delta A = \Delta x \Delta y$, and N_b bins total.

1.2 Parameters

Science signal. The target quantity is the differential phase signal injected into AI $z = 100$:

$$\delta\phi_i(\beta) = A_s \sin(2\pi f t_i) + A_c \cos(2\pi f t_i), \quad \beta = (A_s, A_c), \quad (1)$$

where the signal frequency f is assumed known a priori, leaving two free parameters. The signal amplitude and phase are $A = \sqrt{A_s^2 + A_c^2}$ and $\varphi = \arctan 2(A_c, A_s)$.

Common laser phase. The common laser phase $\phi_i \sim \text{Uniform}(0, 2\pi)$ is shared by both AIs within each shot, drawn independently across shots, and unobserved. It is a latent variable; how we handle it in the likelihood is discussed in Section 5.

Effective AI phase. The total phase entering the transition probability for AI z in shot i is

$$\Phi_{iz}(\phi_i, \beta) = \phi_i + \psi_z(\delta\phi_i(\beta)), \quad \psi_0 = 0, \quad \psi_{100} = \delta\phi_i(\beta). \quad (2)$$

Cloud nuisance parameters. The initial atomic cloud for AI z in shot i is characterised by a Gaussian phase-space distribution with parameters

$$\eta_{iz} = (\mu_{x0}, \mu_{y0}, \mu_{vx}, \mu_{vy}, \sigma_x, \sigma_y, \sigma_{vx}, \sigma_{vy})_{iz}, \quad (3)$$

the transverse centre-of-mass (COM) positions, COM velocities, and Gaussian widths. These vary shot-to-shot and across AIs. Like ϕ_i , they are unobserved and must be handled carefully in the likelihood; see Section 5.

2 Semiclassical Phase-Space Map (PSMAP)

2.1 PSMAP Surrogate

The semiclassical simulation (ais++) integrates the interferometer sequence for a regular 4D grid of initial transverse phase-space coordinates $\mathbf{q}_0 = (x_0, y_0, v_{x0}, v_{y0})$ at fixed source height and longitudinal velocity. For each grid point and each output port p it stores: (i) wavepacket path amplitudes $a_{0,p}(\mathbf{q}_0)$ and $a_{1,p}(\mathbf{q}_0)$; (ii) accumulated phase difference $\Delta\phi_p(\mathbf{q}_0)$; (iii) interference flag $\iota_p \in \{0, 1\}$; and (iv) output state label $s_p \in \{g, e\}$. At runtime, values are obtained by quadrilinear interpolation on the 21^4 grid (PSMAPSurrogate). A separate PSMAP is computed for each AI.

2.2 Transition Probabilities

The probability that an atom with initial condition $\mathbf{q}_0$ is detected at port p given total AI phase Φ is

$$P_p(\mathbf{q}_0; \Phi) = a_{0,p}^2 + a_{1,p}^2 + \iota_p \cdot 2 a_{0,p} a_{1,p} \cos(\Delta\phi_p(\mathbf{q}_0) + \Phi). \quad (4)$$

The per-state probabilities are obtained by summing over ports with matching state label:

$$p_s(\mathbf{q}_0; \Phi) = \sum_{p: s_p=s} P_p(\mathbf{q}_0; \Phi), \quad s \in \{g, e\}. \quad (5)$$

Because the simulation retains only the two dominant interferometry paths,

$$p_g(\mathbf{q}_0; \Phi) + p_e(\mathbf{q}_0; \Phi) \leq 1; \quad (6)$$

atoms not accounted for are lost to pruned paths. In particular, $p_e \neq 1 - p_g$ in general; both are computed independently.

2.3 Ballistic Trajectory Map

Under free-flight propagation over detection time T_{det} , the final transverse position is

$$\mathbf{x}^f(\mathbf{q}_0) = \begin{pmatrix} x_0 + T_{\text{det}} v_{x0} \\ y_0 + T_{\text{det}} v_{y0} \end{pmatrix}. \quad (7)$$

Both output states follow the same trajectory (no state-dependent force before imaging). With default parameters ($T_{\text{det}} = 3.8 \text{ s}$, $\sigma_x = 100 \mu\text{m}$, $\sigma_{v_x} = 309 \mu\text{m/s}$), the cloud expands to $\sigma_{x^f} \approx 1.2 \text{ mm}$ at detection.

3 Per-Pixel Statistics via QMC

3.1 ACS Decomposition

Expanding the cosine in (4), the phase dependence of p_s is entirely captured by three Φ -independent scalars per pixel b , AI z , state s :

$$A_{zsb}(\eta_{iz}) = \mathbb{E}_{\mathbf{q}_0 \sim \text{cloud}, \mathbf{x}^f \in b} [a_0^2(\mathbf{q}_0) + a_1^2(\mathbf{q}_0)], \quad (8)$$

$$C_{zsb}^c(\eta_{iz}) = \mathbb{E}_{\mathbf{q}_0 \sim \text{cloud}, \mathbf{x}^f \in b} [2 \iota a_0(\mathbf{q}_0) a_1(\mathbf{q}_0) \cos \Delta\phi(\mathbf{q}_0)], \quad (9)$$

$$C_{zsb}^s(\eta_{iz}) = \mathbb{E}_{\mathbf{q}_0 \sim \text{cloud}, \mathbf{x}^f \in b} [-2 \iota a_0(\mathbf{q}_0) a_1(\mathbf{q}_0) \sin \Delta\phi(\mathbf{q}_0)], \quad (10)$$

where the expectation is over initial conditions $\mathbf{q}_0$ drawn from the Gaussian cloud $\mathcal{N}(\eta_{iz})$ whose final positions $\mathbf{x}^f(\mathbf{q}_0)$ fall within pixel b . The expected pixel count is then

$$\boxed{\lambda_{izsb}(\Phi) = L_{iz} [A_{zsb} + C_{zsb}^c \cos \Phi + C_{zsb}^s \sin \Phi]}, \quad (11)$$

where $\Phi = \Phi_{iz}(\phi_i, \beta)$ from (2) and L_{iz} is an atom-number scale factor (Section 4.2).

3.2 QMC Evaluation (SurrogatePixelACS)

The conditional expectations (8)–(10) are evaluated by quasi-Monte Carlo (QMC) quadrature with n_{quad} Sobol points.

1. Draw n_{quad} quasi-random standard-normal deviates $\mathbf{z}_j \in \mathbb{R}^4$, $j = 1, \dots, n_{\text{quad}}$, using a scrambled Sobol sequence. These are fixed at construction time and reused across all θ evaluations.

2. Transform to cloud Gaussian: $\mathbf{q}_{0,j} = \mu(\eta) + \text{diag}(\sigma(\eta)) \mathbf{z}_j$.
3. Compute detection positions $\mathbf{x}_j^f$ via (7) and assign each sample to pixel b_j using pixel edge arrays. Samples outside the image boundary are discarded.
4. Evaluate the PSMAP at all $\mathbf{q}_{0,j}$ simultaneously via vectorised quadrilinear interpolation on the `PSMAPSurrogate` grid. This yields amplitudes $(a_{0,jp}, a_{1,jp})$ and phase residuals $\Delta\phi_{jp}$ for all ports p , in one GPU kernel with no Python loops.
5. Read off the ACS components directly from the PSMAP outputs:

$$A_{jp} = a_{0,jp}^2 + a_{1,jp}^2, \quad C_{jp}^c = \iota_p 2 a_{0,jp} a_{1,jp} \cos \Delta\phi_{jp}, \quad C_{jp}^s = -\iota_p 2 a_{0,jp} a_{1,jp} \sin \Delta\phi_{jp}. \quad (12)$$

Sum over ports with matching state label to obtain (A_j, C_j^c, C_j^s) per sample per state.

6. Accumulate per-pixel ACS by equal-weight bincount:

$$A_{zsb} = \frac{1}{n_{\text{in}}} \sum_{j: b_j=b} A_j, \quad (13)$$

and analogously for C_{zsb}^c, C_{zsb}^s , where n_{in} is the number of samples inside the detector. Both ground and excited state statistics are accumulated simultaneously.

The default $n_{\text{quad}} = 20,000$ Sobol points (rounded up to the next power of 2) gives ACS tensors with sub-percent statistical noise in occupied pixels, at a cost of one vectorised PSMAP evaluation per call.

4 Generative Model and Likelihood

4.1 Poisson Observation Model

Conditional on ϕ_i and η_{iz} , pixel counts are independent Poisson:

$$n_{izsb} \sim \text{Poisson}(\lambda_{izsb}(\Phi_{iz})), \quad s \in \{g, e\}, \quad b = 1, \dots, N_b. \quad (14)$$

The per-shot conditional log-likelihood (for both AIs jointly) is

$$\log \mathcal{L}_i(\beta, \phi_i, \{\eta_{iz}\}) = \sum_{z,s,b} \left[n_{izsb} \log \lambda_{izsb}(\Phi_{iz}) - \lambda_{izsb}(\Phi_{iz}) \right], \quad (15)$$

where $\log(n!)$ constants are dropped as parameter-free.

4.2 Atom Number Scale Factor

Rather than tracking the launched atom count Λ_{iz} explicitly, we absorb it into a per-shot, per-AI scale factor L_{iz} defined by the requirement that predicted total counts match observed total counts at the profiled cloud parameters $\hat{\eta}_{iz}$:

$$L_{iz} = \frac{\sum_b (n_{izgb} + n_{izeb})}{\sum_b (A_{zsb}(\hat{\eta}_{iz}) + A_{zsb}^{(e)}(\hat{\eta}_{iz}))}, \quad (16)$$

where the denominator sums the ϕ -independent terms over all pixels for both states. This normalization is exact whenever the interference terms C^c and C^s integrate to zero over all pixels (complementarity), which holds to high accuracy in practice. L_{iz} is computed once at $\hat{\eta}_{iz}$ and treated as fixed.

5 Handling Unobserved Parameters: Profiling and Marginalisation

The full likelihood (15) involves two kinds of unobserved parameters per shot: the common laser phase ϕ_i , and the cloud parameters η_{iz} . Both must be eliminated before optimising over β .

5.1 Marginalise ϕ_i

$\phi_i \sim \text{Uniform}(0, 2\pi)$ is completely unknown. Profiling (jointly optimising ϕ_i with β) would allow the per-shot phase to absorb the signal entirely, biasing $\hat{\beta}$ toward zero. We therefore marginalise numerically using a uniform phase grid $\phi_k = 2\pi k/K$, $k = 0, \dots, K-1$:

$$\log \mathcal{L}_i^{\text{marg}}(\beta, \{\hat{\eta}_{iz}\}) = \text{logsumexp}_{k=0}^{K-1} [\log \mathcal{L}_i(\beta, \phi_k, \{\hat{\eta}_{iz}\})] - \log K. \quad (17)$$

This approximates the exact integral over ϕ_i to arbitrary accuracy as $K \rightarrow \infty$. The required grid size scales as

$$K \gtrsim 2\sqrt{\bar{N}}, \quad (18)$$

where $\bar{N}$ is the mean detected atom count per shot per AI. In practice $K = 512$ is used.

5.2 Profile η_{iz}

Each η_{iz} has 8 components; for a 32×32 image, there are 2048 Poisson measurements per shot per AI. The likelihood is extremely sharply peaked in η_{iz} , so profiling and marginalising are numerically equivalent. We profile: for each shot i and AI z , find

$$\hat{\eta}_{iz} = \arg \max_{\eta} \left[\text{logsumexp}_k \log \mathcal{L}_i^{(z)}(\eta, \phi_k) + \log p(\eta) \right], \quad (19)$$

where $\log \mathcal{L}_i^{(z)}$ is the single-AI, single-shot conditional logL using *both* ports jointly:

$$\log \mathcal{L}_i^{(z)}(\eta, \phi_k) = \sum_{s \in \{g, e\}} \sum_b \left[n_{izsb} \log \lambda_{izsb}(\phi_k; \eta) - \lambda_{izsb}(\phi_k; \eta) \right], \quad (20)$$

and $\log p(\eta)$ is a Gaussian prior (Section 5.3).

Crucially, this profiling objective already marginalises over ϕ_i via logsumexp , using the same phase grid as the signal fit. The resulting $\hat{\eta}_{iz}$ is then fixed for all subsequent evaluations of $\log \mathcal{L}(\beta)$.

Why $\hat{\eta}$ is approximately independent of β . The cloud parameters η_{iz} control the spatial distribution of atoms across pixels; β controls the port fractions via the fringe. These two effects are largely orthogonal, so $\hat{\eta}_{iz}(\beta)$ changes negligibly as β varies. This justifies computing $\hat{\eta}_{iz}$ once (at arbitrary β , e.g. $\beta = 0$) and treating it as fixed during the outer β optimisation.

5.3 Prior on Cloud Parameters

The Gaussian log-prior on η_{iz} is

$$\begin{aligned} \log p(\eta) = & -\frac{\mu_{x_0}^2}{2\sigma_{\mu, \text{pos}}^2} - \frac{\mu_{y_0}^2}{2\sigma_{\mu, \text{pos}}^2} - \frac{\mu_{v_x}^2}{2\sigma_{\mu, \text{vel}}^2} - \frac{\mu_{v_y}^2}{2\sigma_{\mu, \text{vel}}^2} \\ & - \frac{(\sigma_x - \bar{\sigma}_{\text{pos}})^2}{2\sigma_{\sigma, \text{pos}}^2} - \frac{(\sigma_y - \bar{\sigma}_{\text{pos}})^2}{2\sigma_{\sigma, \text{pos}}^2} - \frac{(\sigma_{v_x} - \bar{\sigma}_{\text{vel}})^2}{2\sigma_{\sigma, \text{vel}}^2} - \frac{(\sigma_{v_y} - \bar{\sigma}_{\text{vel}})^2}{2\sigma_{\sigma, \text{vel}}^2}, \end{aligned} \quad (21)$$

where the hyperparameters $(\sigma_{\mu,\text{pos}}, \sigma_{\mu,\text{vel}}, \bar{\sigma}_{\text{pos}}, \bar{\sigma}_{\text{vel}}, \sigma_{\sigma,\text{pos}}, \sigma_{\sigma,\text{vel}})$ are parsed from the dataset name and can be overridden by command-line arguments.

The optimisation uses a log-sigma parameterisation of the cloud widths: $\tilde{\sigma}_x = \log \sigma_x$ etc., so that positivity of widths is enforced by the domain of $\mathbb{R}$ without explicit bounds. Bounded L-BFGS-B (rather than Nelder–Mead) is used, with bounds on the log-widths derived from the physical constraints of the image field of view.

Initialisation. Two strategies are supported:

- *Oracle init:* $\hat{\eta}_{iz}^{(0)} = \text{true cloud parameters read from the H5 file. Used for validation; referred to as } \text{--init-from-true}.$
- *Prior-mean init:* $\hat{\eta}_{iz}^{(0)} = (0, 0, 0, 0, \bar{\sigma}_{\text{pos}}, \bar{\sigma}_{\text{pos}}, \bar{\sigma}_{\text{vel}}, \bar{\sigma}_{\text{vel}}).$

6 Alternative: Moment-Based Cloud Marginalisation

6.1 Motivation

The profiling strategy of Section 5 finds a single $\hat{\eta}_{iz}$ per shot and fixes it. This is valid when the likelihood is sharply peaked in η , but empirical multi-start optimisation (Section 9) reveals that the likelihood surface has extended *ridges* in η -space, arising from ballistic degeneracies (Section 6.2). Fixing $\hat{\eta}$ at one point on a ridge introduces a systematic error in the marginalised signal posterior.

This section develops an alternative that replaces the per-pixel likelihood over all N_b bins with a likelihood over a small set of image *moments*, enabling an analytic posterior $p(\eta|m)$ that concentrates mass on the ridge and can be used as a proposal or prior for a proper marginalisation over η .

6.2 Ballistic Degeneracies and Ridge Structure

Under free-flight propagation (7) the final-plane Gaussian parameters are related to the initial ones by

$$\mu_{xf} = \mu_{x0} + T_{\text{det}} \mu_{vx}, \quad (22)$$

$$\sigma_{xf}^2 = \sigma_{x0}^2 + T_{\text{det}}^2 \sigma_{vx}^2, \quad (23)$$

and identically for y . The cloud image is predominantly determined by the four *final* moments $(\mu_{xf}, \mu_{yf}, \sigma_{xf}, \sigma_{yf})$. Each of (22)–(23) is a constraint relating two initial parameters to one observable, leaving a one-dimensional family of initial conditions consistent with any given image. In (μ_{x0}, μ_{vx}) space this is the line $\mu_{x0} + T_{\text{det}} \mu_{vx} = \mu_{xf}$; in $(\sigma_{x0}, \sigma_{vx})$ space it is the ellipse $\sigma_{x0}^2 + T_{\text{det}}^2 \sigma_{vx}^2 = \sigma_{xf}^2$. These are precisely the ridges observed empirically.

6.3 Summary Statistics and the Key Decomposition

Total-count image. Define the per-pixel sum over output states

$$m_b^{(iz)} = n_{izgb} + n_{izeb}, \quad (24)$$

and define the scalar image moments

$$N_z = \sum_b m_b^{(iz)}, \quad (25)$$

$$\hat{\mu}_{xf} = N_z^{-1} \sum_b x_b m_b^{(iz)}, \quad (26)$$

$$\hat{\sigma}_{xf}^2 = N_z^{-1} \sum_b (x_b - \hat{\mu}_{xf})^2 m_b^{(iz)}, \quad (27)$$

and analogously $\hat{\mu}_{yf}, \hat{\sigma}_{yf}^2$. We collect these into $m = (N_z, \hat{\mu}_{xf}, \hat{\sigma}_{xf}^2, \hat{\mu}_{yf}, \hat{\sigma}_{yf}^2)$.

Science parameters do not enter m . The expected total-count rate at pixel b is

$$\mathbb{E}[m_b^{(iz)}] = L_{iz} [(A_{zgb} + A_{zeb}) + (C_{zgb}^c + C_{zeb}^c) \cos \Phi + (C_{zgb}^s + C_{zeb}^s) \sin \Phi]. \quad (28)$$

In a balanced two-path interferometer the output ports are complementary: atoms transmitted into the ground port are not transmitted into the excited port, so $A_{zgb} + A_{zeb} \approx \text{const}$ and

$$C_{zgb}^c + C_{zeb}^c \approx 0, \quad C_{zgb}^s + C_{zeb}^s \approx 0. \quad (29)$$

Therefore $\mathbb{E}[m_b^{(iz)}] \approx L_{iz}(A_{zgb} + A_{zeb})$, independent of the total AI phase Φ and hence independent of both the laser phase ϕ_i and the science signal $\beta = (A_s, A_c)$. Consequently,

$$\boxed{p(m \mid \eta, \beta, \phi_i) = p(m \mid \eta)} \quad (30)$$

exactly to the extent that (29) holds. The moments are a sufficient statistic for η that carries *zero* information about β . This decomposition is not an approximation: the science signal $\delta\phi_i(\beta)$ and the laser phase ϕ_i both enter the image exclusively through the interference terms $C^c \cos \Phi + C^s \sin \Phi$, which cancel in the port sum.

6.4 Moment Likelihood $p(m \mid \eta)$

Under (29) the total-count image is a 2D Poisson process with pixel rates $\lambda_b = L_{iz} A_{zb}^{tot}(\eta)$, where $A_{zb}^{tot} = A_{zgb} + A_{zeb}$. For the cloud model of Section 5, $A_{zb}^{tot}(\eta)$ is proportional to the probability that a Gaussian-distributed atom lands in pixel b at detection time.

By the central limit theorem for large N_z , the sample moments converge to

$$\hat{\mu}_{xf} \mid \eta \sim \mathcal{N}\left(\mu_{xf}(\eta), \frac{\sigma_{xf}^2(\eta)}{N_z}\right), \quad (31)$$

$$\hat{\sigma}_{xf}^2 \mid \eta \sim \mathcal{N}\left(\sigma_{xf}^2(\eta), \frac{2\sigma_{xf}^4(\eta)}{N_z}\right), \quad (32)$$

and identically for y . All four are independent. The forward model is (22)–(23) with N_z known from the observed total count and σ_{xf}^2 estimated from the data (replacing the true value in the noise variance by its estimate introduces a negligible bias at large N_z).

Assumption: PSMAP undetectable in image moments. Equation (29) is exact when port probabilities sum to one. In practice, paths are pruned (Section 2.2), so the fringe terms do not cancel perfectly. The residual fringe contrast is $\epsilon_{\text{fringe}} = |C_{zb}^c + C_{z(e)b}^c|/(A_{zb}^{tot}) \ll 1$. Furthermore, the spatially varying phase $\Delta\phi_p(\mathbf{q}_0)$ (the PSMAP contribution) averages over the cloud and partially cancels in the moment integrals. We assume this residual is negligible compared with the Poisson uncertainty $1/\sqrt{N_z}$ on the moments, i.e. the PSMAP fringe structure is undetectable in the image given the available atom number. Under this assumption (30) holds exactly and the moment-based approach is not an approximation but an alternative sufficient statistic.

6.5 Analytic Posterior $p(\eta \mid m)$

6.5.1 Mean Parameters

Consider the x -direction means $\theta_\mu = (\mu_{x_0}, \mu_{v_x})^\top$. The observation model (31) combined with the prior (21) takes the form of a linear Gaussian model:

$$\hat{\mu}_{xf} = \mathbf{h}^\top \theta_\mu + \epsilon, \quad \mathbf{h} = \begin{pmatrix} 1 \\ T_{\text{det}} \end{pmatrix}, \quad \epsilon \sim \mathcal{N}(0, R_x), \quad R_x = \frac{\hat{\sigma}_{xf}^2}{N_z}, \quad (33)$$

with Gaussian prior $\theta_\mu \sim \mathcal{N}(\mathbf{0}, \mathbf{P}_0)$, $\mathbf{P}_0 = \text{diag}(\sigma_{\mu,\text{pos}}^2, \sigma_{\mu,\text{vel}}^2)$. The posterior is analytic (standard Bayesian linear regression / Kalman update):

$$\mathbf{K} = \mathbf{P}_0 \mathbf{h} (\mathbf{h}^\top \mathbf{P}_0 \mathbf{h} + R_x)^{-1}, \quad (34)$$

$$\boldsymbol{\mu}_{\text{post}} = \mathbf{K} \hat{\boldsymbol{\mu}}_{xf}, \quad (35)$$

$$\mathbf{P}_{\text{post}} = (\mathbf{I} - \mathbf{K} \mathbf{h}^\top) \mathbf{P}_0. \quad (36)$$

The posterior $\theta_\mu \mid \hat{\boldsymbol{\mu}}_{xf} \sim \mathcal{N}(\boldsymbol{\mu}_{\text{post}}, \mathbf{P}_{\text{post}})$ is a 2D Gaussian with $\mathbf{P}_{\text{post}}$ having one very small eigenvalue (the direction pinned by the constraint $\mu_{x_0} + T_{\text{det}} \mu_{v_x} = \hat{\boldsymbol{\mu}}_{xf}$) and one larger eigenvalue (the unconstrained direction along the ridge). An identical update applies to (μ_{y_0}, μ_{v_y}) using $\hat{\boldsymbol{\mu}}_{yf}$.

6.5.2 Width Parameters

Define the squared widths $s_x = \sigma_{x_0}^2$, $s_{v_x} = \sigma_{v_x}^2$. The observation model (32) becomes

$$\hat{\sigma}_{xf}^2 = s_x + T_{\text{det}}^2 s_{v_x} + \epsilon_\sigma, \quad \epsilon_\sigma \sim \mathcal{N}(0, R_\sigma), \quad R_\sigma = \frac{2 \hat{\sigma}_{xf}^4}{N_z}. \quad (37)$$

This is linear in (s_x, s_{v_x}) , so the Kalman update (34)–(36) applies with $\mathbf{h} = (1, T_{\text{det}}^2)^\top$ and prior variance on (s_x, s_{v_x}) derived from (21) by the delta method. The non-negativity constraints $s_x, s_{v_x} \geq 0$ truncate the posterior; the truncation is negligible when the prior is well away from zero.

Dominant-velocity-spread regime. In the MAGIS-100 setup ($T_{\text{det}} = 3.8 \text{ s}$, $\sigma_{v_x} \sim 100\text{--}309 \mu\text{m/s}$), $T_{\text{det}} \sigma_{v_x} \sim 380\text{--}1170 \mu\text{m} \gg \sigma_{x_0} \sim 100 \mu\text{m}$. Expanding (23) to first order in σ_{x_0}/σ_{xf} :

$$\sigma_{xf} \approx T_{\text{det}} \sigma_{v_x} + \frac{\sigma_{x_0}^2}{2 T_{\text{det}} \sigma_{v_x}}. \quad (38)$$

To leading order $\sigma_{v_x} \approx \hat{\sigma}_{xf}/T_{\text{det}}$ is directly observed, while σ_{x_0} enters only as a small correction and is poorly constrained by the image alone. The posterior on σ_{v_x} is therefore narrow ($\propto$ width of the Kalman update (36)), while the posterior on σ_{x_0} remains broad, reflecting its unidentifiability from a single-epoch image.

6.6 Marginalization via the Moment Posterior

The moment posterior factorises as

$$p(\eta \mid m) = p(\mu_{x_0}, \mu_{v_x} \mid \hat{\boldsymbol{\mu}}_{xf}) \cdot p(\mu_{y_0}, \mu_{v_y} \mid \hat{\boldsymbol{\mu}}_{yf}) \cdot p(\sigma_{x_0}^2, \sigma_{v_x}^2 \mid \hat{\sigma}_{xf}^2) \cdot p(\sigma_{y_0}^2, \sigma_{v_y}^2 \mid \hat{\sigma}_{yf}^2), \quad (39)$$

each factor being an analytic Gaussian (or truncated Gaussian) derived above.

The signal posterior is then

$$p(\beta \mid \text{data}) \propto \int d\eta p(\text{data} \mid \beta, \eta) p(\eta \mid m). \quad (40)$$

This integral has two key advantages over the full marginalisation with prior $p(\eta)$:

1. **Constrained support.** $p(\eta \mid m)$ concentrates mass on a 4-dimensional manifold (the four ballistic ridges) rather than spanning the full 8D prior volume. The effective sample size required for accurate Monte Carlo integration is thus dramatically reduced.
2. **Analytic sampler.** Samples $\eta_i \sim p(\eta \mid m)$ are free: draw from four independent 2D Gaussians. No PSMAP evaluation is needed to generate starting points.

Double-counting note. The moments m are computed from the same data used in $p(\text{data}|\beta, \eta)$. Double-counting is avoided because (30) guarantees that m carries no information about β : the moment channel is orthogonal to the signal channel. The only overlap is between m and the cloud channel of $p(\text{data}|\beta, \eta)$ — but this overlap vanishes by the assumption that the PSMAP fringe is undetectable in the moments. When this assumption is not exact, one should use importance sampling to correct for it (see Section 6.8).

6.7 Analytical $R = 0$ Limit: Dimension Reduction to a Point Estimate

A practically important special case arises when we treat the sample moments as *exact* observations — i.e. set the measurement noise to zero ($R_x = R_\sigma = 0$ in (33)–(37)). This is justified when $N_z \gg 1$ so that the CLT variance of the sample moments is negligible relative to the prior spread.

$R=0$ Kalman posterior. Setting $R_x \rightarrow 0$, the Kalman gain in (34) satisfies

$$\mathbf{K} \rightarrow \frac{\mathbf{P}_0 \mathbf{h}}{\mathbf{h}^\top \mathbf{P}_0 \mathbf{h}},$$

and the posterior covariance $\mathbf{P}_{\text{post}} \rightarrow \mathbf{P}_0 - \mathbf{P}_0 \mathbf{h} \mathbf{h}^\top \mathbf{P}_0 / (\mathbf{h}^\top \mathbf{P}_0 \mathbf{h})$ is *rank-deficient*: it has a zero eigenvalue in the $\mathbf{h}$ direction. The posterior is a degenerate Gaussian — a delta function in the constraint direction $\mathbf{h}^\top \theta = z$ (the ballistic manifold), and a 1D Gaussian in the orthogonal (ridge) direction.

Dimension reduction. Each of the four 2D blocks (μ_{x_0}, μ_{v_x}) , (μ_{y_0}, μ_{v_y}) , $(\sigma_{x_0}^2, \sigma_{v_x}^2)$, $(\sigma_{y_0}^2, \sigma_{v_y}^2)$ has one degree of freedom eliminated by its $R=0$ constraint. The full 8-dimensional nuisance space $\eta \in \mathbb{R}^8$ collapses to a **4-dimensional** product of ridge manifolds:

$$\{\eta : \mu_{x_0} + T\mu_{v_x} = \hat{\mu}_{xf}\} \times \{\eta : \mu_{y_0} + T\mu_{v_y} = \hat{\mu}_{yf}\} \times \{\eta : \sigma_{x_0}^2 + T^2\sigma_{v_x}^2 = \hat{\sigma}_{xf}^2\} \times \{\eta : \sigma_{y_0}^2 + T^2\sigma_{v_y}^2 = \hat{\sigma}_{yf}^2\}. \quad (41)$$

The marginal (40) thus reduces to a *4D integral* over these four ridges.

Point-estimate approximation (zero integration). If we further approximate the marginal by evaluating the likelihood at a *single* representative point per ridge — the posterior mean, which is the minimum-prior-cost point on each constraint manifold — the integral reduces to a scalar evaluation. For the mean block with diagonal prior $P_0 = \text{diag}(\tau_{\text{pos}}^2, \tau_{\text{vel}}^2)$ and measurement $\hat{\mu}_{xf}$ (exact), this gives:

$$\hat{\mu}_{x_0} = \frac{\tau_{\text{pos}}^2}{\tau_{\text{pos}}^2 + T^2 \tau_{\text{vel}}^2} \hat{\mu}_{xf}, \quad \hat{\mu}_{v_x} = \frac{T \tau_{\text{vel}}^2}{\tau_{\text{pos}}^2 + T^2 \tau_{\text{vel}}^2} \hat{\mu}_{xf}, \quad (42)$$

and analogously for y and the variance blocks. No integration or Monte Carlo is required.

Bias and validity of the point estimate. The true η lies on the constraint manifold (41) but in general not at the minimum-norm point (42). Using the point estimate therefore introduces a bias in the ACS evaluation unless the fringe coefficients are constant along the ridge. Section 7 describes the unbiased 4D integral implemented in `python-scripts/moment_inference.py`.

6.8 Importance Sampling Correction and Information Diagnostic

If the PSMAP fringe is not fully cancelled in the moment integrals, using $p(\eta|m)$ as a prior introduces a small bias. This is corrected exactly by importance sampling:

1. Draw M samples $\eta_j \sim q(\eta) = p(\eta \mid m_\alpha)$, where m_α is the moment posterior with covariances inflated by a factor $\alpha > 1$ (e.g. $\alpha = 4$) to ensure the proposal has heavier tails than the true posterior.
2. Evaluate the unnormalised importance weights

$$w_j = \frac{p(\eta_j)}{q(\eta_j)} \propto \frac{p(\eta_j)}{p(\eta_j \mid m_\alpha)}, \quad (43)$$

which correct for using q in place of the flat prior $p(\eta)$.

3. Approximate the integral (40) by

$$p(\beta \mid \text{data}) \propto \sum_{j=1}^M w_j p(\text{data} \mid \beta, \eta_j). \quad (44)$$

Information diagnostic. The effective sample size

$$\text{ESS} = \frac{(\sum_j w_j)^2}{\sum_j w_j^2} \quad (45)$$

measures how much the full pixel likelihood reshapes $p(\eta)$ relative to $p(\eta \mid m)$. If $\text{ESS}/M \approx 1$, the moment posterior $p(\eta \mid m)$ is essentially the correct posterior — the PSMAP fringe structure adds no detectable information about η beyond the cloud envelope moments, confirming the assumption of Section 6. If $\text{ESS}/M \ll 1$, the PSMAP contributes additional cloud information and full joint inference is required.

7 4D Gauss-Hermite Moment Inference

This section describes the complete, unbiased implementation of the $R=0$ ridge integral introduced in Section 6.7. Rather than fixing η at the single minimum-norm ridge point, we integrate over all initial conditions consistent with the observed final moments using Gauss-Hermite quadrature with an analytic weight function derived from the $R=0$ posterior. No per-pixel spatial information is used; the sufficient statistic is reduced to five scalars per shot per AI.

7.1 Spatially-Integrated ACS

Instead of the per-pixel ACS tensors $\{A_{zsb}, C_{zsb}^c, C_{zsb}^s\}$ of Section 3, we work with their spatial sums,

$$A_{zs}(\eta) = \sum_b A_{zsb}(\eta), \quad C_{zs}^c(\eta) = \sum_b C_{zsb}^c(\eta), \quad C_{zs}^s(\eta) = \sum_b C_{zsb}^s(\eta), \quad (46)$$

reducing the cloud-marginalized pixel ACS to three scalars per state per AI. These are the expectations of the PSMAP amplitudes over the full cloud Gaussian, without conditioning on final position:

$$A_{zs}(\eta) = \mathbb{E}_{\mathbf{q}_0 \sim \mathcal{N}(\eta)} \left[\sum_{p: s_p=s} (a_{0,p}^2(\mathbf{q}_0) + a_{1,p}^2(\mathbf{q}_0)) \right], \quad (47)$$

$$C_{zs}^c(\eta) = \mathbb{E}_{\mathbf{q}_0 \sim \mathcal{N}(\eta)} \left[\sum_{p: s_p=s} \iota_p 2 a_{0,p}(\mathbf{q}_0) a_{1,p}(\mathbf{q}_0) \cos \Delta \phi_p(\mathbf{q}_0) \right], \quad (48)$$

$$C_{zs}^s(\eta) = \mathbb{E}_{\mathbf{q}_0 \sim \mathcal{N}(\eta)} \left[- \sum_{p: s_p=s} \iota_p 2 a_{0,p}(\mathbf{q}_0) a_{1,p}(\mathbf{q}_0) \sin \Delta \phi_p(\mathbf{q}_0) \right]. \quad (49)$$

The expected detected count for state s at total phase Φ is then

$$\lambda_{zs}(\Phi; \eta) = \Lambda_{iz} [A_{zs}(\eta) + C_{zs}^c(\eta) \cos \Phi + C_{zs}^s(\eta) \sin \Phi], \quad (50)$$

where Λ_{iz} is an atom-count normalisation factor defined below.

Monte Carlo evaluation. Let n_{qmc} standard-normal deviates $\mathbf{z}_j \in \mathbb{R}^4$ be drawn once. Given η , the cloud samples are $\mathbf{q}_{0,j} = \mu(\eta) + \text{diag}(\sigma(\eta)) \mathbf{z}_j$. Evaluate the PSMAP at all n_{qmc} points in a single batched GPU call, then average. Because the samples are not conditioned on final position, the estimator variance is $\mathcal{O}(1/n_{\text{qmc}})$ without the $1/n_{\text{occupied}}$ binning penalty of the per-pixel estimator. The default $n_{\text{qmc}} = 2048$ is sufficient because the required ACS accuracy is set by per-shot signal-to-noise, not per-pixel count statistics.

7.2 Port-Fraction Likelihood

The observations per shot per AI are reduced to two scalars: total ground-state count $n_{izg} = \sum_b n_{izgb}$ and total excited-state count $n_{ize} = \sum_b n_{izeb}$, with $N_{iz} = n_{izg} + n_{ize}$. Define

$$A_z^{\text{tot}} = A_{zg} + A_{ze}, \quad C_z^{c,\text{tot}} = C_{zg}^c + C_{ze}^c, \quad C_z^{s,\text{tot}} = C_{zg}^s + C_{ze}^s. \quad (51)$$

Because path-pruning violates complementarity ($C_z^{c,\text{tot}} \neq 0$ in general), the total expected count at phase Φ is Φ -dependent:

$$\lambda_{iz}^{\text{tot}}(\Phi; \eta) = \Lambda_{iz} [A_z^{\text{tot}} + C_z^{c,\text{tot}} \cos \Phi + C_z^{s,\text{tot}} \sin \Phi]. \quad (52)$$

We normalise by choosing $\Lambda_{iz}(\Phi; \eta)$ so that the predicted total matches the observed total at every phase:

$$\Lambda_{iz}(\Phi; \eta) = \frac{N_{iz}}{A_z^{\text{tot}}(\eta) + C_z^{c,\text{tot}}(\eta) \cos \Phi + C_z^{s,\text{tot}}(\eta) \sin \Phi}. \quad (53)$$

This Φ -dependent normalisation eliminates the total-count channel, leaving the log-likelihood as a function of the port fraction only:

$$\hat{\lambda}_{izg}(\Phi; \eta) = N_{iz} \frac{A_{zg} + C_{zg}^c \cos \Phi + C_{zg}^s \sin \Phi}{A_z^{\text{tot}} + C_z^{c,\text{tot}} \cos \Phi + C_z^{s,\text{tot}} \sin \Phi}, \quad (54)$$

$$\hat{\lambda}_{ize}(\Phi; \eta) = N_{iz} - \hat{\lambda}_{izg}(\Phi; \eta). \quad (55)$$

The Poisson log-likelihood for AI z at phase Φ is

$$\ell_{iz}(\Phi; \eta) = n_{izg} \log \hat{\lambda}_{izg} - \hat{\lambda}_{izg} + n_{ize} \log \hat{\lambda}_{ize} - \hat{\lambda}_{ize}, \quad (56)$$

which depends only on the port fraction $p_g(\Phi; \eta) = \hat{\lambda}_{izg}/N_{iz}$. The combined Z0+Z100 conditional log-likelihood for shot i at laser phase ϕ is

$$\ell_i(\phi; \eta_{iz0}, \eta_{iz100}, \beta) = \ell_{iz0}(\phi; \eta_{iz0}) + \ell_{iz100}(\phi + \delta\phi_i(\beta); \eta_{iz100}), \quad (57)$$

where $\delta\phi_i(\beta) = A_s \sin(2\pi f i) + A_c \cos(2\pi f i)$ is the differential phase signal injected into Z100.

7.3 1D Ridge Prior from the R=0 Kalman Update

Under the R=0 Kalman update (Section 6.7), the posterior for the mean block (μ_{x0}, μ_{vx}) conditional on $\hat{\mu}_{xf}$ is a degenerate Gaussian with a non-degenerate 1D marginal in the direction parametrised by μ_{x0} . Substituting the R=0 Kalman gain gives

$$\mu_{x0} \mid \hat{\mu}_{xf} \sim \mathcal{N}(\bar{\mu}_{x0}, \bar{\sigma}_{\mu_{x0}}^2), \quad (58)$$

$$\bar{\mu}_{x0} = \frac{\tau_{\text{pos}}^2}{\tau_{\text{pos}}^2 + T^2 \tau_{\text{vel}}^2} \hat{\mu}_{xf}, \quad \bar{\sigma}_{\mu_{x0}}^2 = \left(\frac{1}{\tau_{\text{pos}}^2} + \frac{1}{T^2 \tau_{\text{vel}}^2} \right)^{-1}. \quad (59)$$

Once μ_{x0} is fixed, the ridge constraint gives $\mu_{vx} = (\hat{\mu}_{xf} - \mu_{x0})/T$. An identical pair of equations applies to (μ_{y0}, μ_{vy}) .

For the variance block, a linearised prior on $s_{x_0} = \sigma_{x_0}^2$, $s_{v_x} = \sigma_{v_x}^2$ and the R=0 Kalman update on the constraint $s_{x_0} + T^2 s_{v_x} = \hat{\sigma}_{x_f}^2$ give

$$s_{x_0} \mid \hat{\sigma}_{x_f}^2 \sim \mathcal{N}(\bar{s}_{x_0}^*, \bar{\tau}_{s_{x_0}}^2), \quad (60)$$

$$\bar{s}_{x_0}^* = \bar{\sigma}_{\text{pos}}^2 + \bar{\tau}_{s_{x_0}}^2 \frac{\hat{\sigma}_{x_f}^2 - \bar{\sigma}_{\text{pos}}^2 - T^2 \bar{\sigma}_{\text{vel}}^2}{\tau_{s_{x_0}}^2}, \quad \bar{\tau}_{s_{x_0}}^2 = \left(\frac{1}{\tau_{s_{x_0}}^2} + \frac{1}{T^4 \tau_{s_{v_x}}^2} \right)^{-1}, \quad (61)$$

where $\tau_{s_{x_0}} = 2\bar{\sigma}_{\text{pos}}\sigma_{\sigma_{\text{pos}}}$ is the prior standard deviation on s_{x_0} obtained by the delta method, and $\bar{\sigma}_{\text{pos}}$, $\sigma_{\sigma_{\text{pos}}}$ are the prior mean and standard deviation of σ_{x_0} . Once s_{x_0} is fixed, $s_{v_x} = \max(\hat{\sigma}_{x_f}^2 - s_{x_0}, 0)/T^2$. These four 1D priors are the analytic Bayesian combination of the original 8D prior $p(\eta)$ with the observed final moments, not a heuristic.

7.4 4D Gauss-Hermite Quadrature Grid

The 4D nuisance integral after the R=0 reduction is

$$\int_{\mathbb{R}^4} \ell_i(\phi; \tilde{\eta}(\xi), \beta) p_{\text{ridge}}(\xi \mid m_i) d^4\xi, \quad (62)$$

where $\xi = (\mu_{x_0}, \mu_{y_0}, s_{x_0}, s_{y_0})$ collects the four free parameters, $\tilde{\eta}(\xi)$ reconstructs the full 8-vector via the ridge constraints, and $p_{\text{ridge}} = \prod_{d=1}^4 \mathcal{N}(\xi_d, \bar{\tau}_d^2)$ is the product of four independent 1D Gaussians from Section 7.3.

Since p_{ridge} is a product of Gaussians, the integral is approximated by a 4D product Gauss-Hermite (GH) rule. With n_{gh} nodes per dimension, the rule uses n_{gh}^4 points and is exact for polynomials of degree $\leq 2n_{\text{gh}} - 1$ in each dimension.

Construction. Let $\{t_k, w_k\}_{k=1}^{n_{\text{gh}}}$ be the standard Hermite nodes and weights ($\sum_k w_k = \sqrt{\pi}$). For dimension d with Gaussian $\mathcal{N}(\bar{\xi}_d, \bar{\tau}_d^2)$, the physical nodes and normalised log-weights are

$$\xi_{d,k} = \bar{\xi}_d + \sqrt{2} \bar{\tau}_d t_k, \quad \log \tilde{w}_{d,k} = \log w_k - \frac{1}{2} \log \pi, \quad (63)$$

with $\sum_k \tilde{w}_{d,k} = 1$. The 4D product rule has node $\xi^{(j)} = (\xi_{1,k_1}, \xi_{2,k_2}, \xi_{3,k_3}, \xi_{4,k_4})$ and

$$\log \tilde{w}^{(j)} = \sum_{d=1}^4 \log \tilde{w}_{d,k_d}. \quad (64)$$

Variance nodes s_{x_0} are clipped to $[0, \hat{\sigma}_{x_f}^2]$ before reconstructing $\tilde{\eta}$:

$$\mu_{v_x} = (\hat{\mu}_{x_f} - \mu_{x_0})/T, \quad \sigma_{v_x} = \sqrt{\max(\hat{\sigma}_{x_f}^2 - s_{x_0}, 0)}/T, \quad (65)$$

and analogously for the y block.

7.5 Per-Shot Log-Marginal Likelihood

For each GH node $j = 1, \dots, n_{\text{gh}}^4$, the integrated ACS $\{A_{zs}^{(j)}, C_{zs}^{c(j)}, C_{zs}^{s(j)}\}$ is evaluated once via the batched Monte Carlo estimator of Section 7.1 at $\tilde{\eta}^{(j)}$, and cached. All subsequent evaluations at different (ϕ, β) reuse these precomputed values.

The GH-marginalised log-likelihood for AI z at phase Φ is

$$\log f_{iz}(\Phi) = \log \text{sumexp}_{j=1}^{n_{\text{gh}}^4} \left[\log \tilde{w}^{(j)} + \ell_{iz}(\Phi; \tilde{\eta}^{(j)}) \right]. \quad (66)$$

The phase-marginalised log-likelihood for shot i is obtained by combining the two AIs and marginalising ϕ_i over a uniform grid $\phi_k = 2\pi k/K$, $k = 0, \dots, K-1$:

$$\log \mathcal{L}_i(\beta) = \text{logsumexp}_{k=0}^{K-1} \left[\log f_{iz_0}(\phi_k) + \log f_{iz_{100}}(\phi_k + \delta\phi_i(\beta)) \right] - \log K. \quad (67)$$

The outer logsumexp marginalises over the unknown laser phase ϕ_i with a uniform discrete prior. The full log-likelihood sums over independent shots:

$$\log \mathcal{L}(\beta) = \sum_{i=1}^{N_{\text{shots}}} \log \mathcal{L}_i(\beta). \quad (68)$$

7.6 Signal Optimisation

The MLE $\hat{\beta} = \arg \max_{\beta} \log \mathcal{L}(\beta)$ is found by multi-start L-BFGS-B. One start is fixed at the null $(0, 0)$; the remaining $n_{\text{starts}} - 1$ are drawn uniformly from the search domain $[-A_{\text{max}}, A_{\text{max}}]^2$. The best solution across all starts is retained. With $N_{\text{shots}} \geq 100$ and signal amplitude $A \gtrsim 0.05$ rad, all starts converge to the same optimum, confirming unimodality at this regime.

7.7 Complete Algorithm

1. **Data reduction** (once per shot per AI). From the 2D image pair (n_{izgb}, n_{izeb}) , compute port sums n_{izg} , n_{ize} , N_{iz} , and spatial moments $(\hat{\mu}_{xf}, \hat{\mu}_{yf}, \hat{\sigma}_{xf}^2, \hat{\sigma}_{yf}^2)$ from the combined image $m_b = n_{izgb} + n_{izeb}$.
2. **Ridge prior** (once per shot per AI). Compute the four 1D Gaussian parameters $\{(\bar{\mu}_{x_0}, \bar{\sigma}_{\mu_{x_0}}), (\bar{\mu}_{y_0}, \bar{\sigma}_{\mu_{y_0}}), (\bar{s}_x, \bar{\sigma}_{s_x}), (\bar{s}_y, \bar{\sigma}_{s_y})\}$ via (59) and (61).
3. **GH grid** (once per shot per AI). Form the n_{gh}^4 product-rule nodes $\{\xi^{(j)}\}$ and log-weights $\{\log \tilde{w}^{(j)}\}$ via (63)–(64). Clip variance nodes and reconstruct $\tilde{\eta}^{(j)}$ via (65).
4. **Integrated ACS precomputation** (once per GH node per shot per AI). For each $\tilde{\eta}^{(j)}$, draw n_{qmc} Gaussian cloud samples and evaluate the batched PSMAP on GPU to obtain $(A_{zs}^{(j)}, C_{zs}^{c(j)}, C_{zs}^{s(j)})$ for $s \in \{g, e\}$. Store alongside $\log \tilde{w}^{(j)}$.
5. **Signal log-likelihood** (per β evaluation). For each shot i and trial β : compute $\delta\phi_i(\beta)$; evaluate $\log f_{iz}(\phi_k)$ for all k via (66); apply (67); sum shots via (68).
6. **Optimisation**. Maximise $\log \mathcal{L}(\beta)$ over $\beta \in [-A_{\text{max}}, A_{\text{max}}]^2$ with n_{starts} independent L-BFGS-B runs (Section 7.6).

7.8 Default Hyperparameters and Computational Cost

7.9 Validation

Running `moment_inference.py` on the $N_{\text{shots}} = 200$, $N_{iz} = 10^6$ dataset (true signal $A = 0.10$ rad, $f = 0.30$ cycles/shot, $\varphi = 0.50$ rad, giving $(A_s, A_c) = (0.0878, 0.0479)$ rad) gives:

- $\log \mathcal{L}(\hat{\beta}) - \log \mathcal{L}(0, 0) = +1.79 \times 10^5$ nats, strongly detecting the signal.
- $\hat{A}_s = 0.091$ rad, $\hat{A}_c = 0.051$ rad (recovered amplitude $\hat{A} = 0.104$ rad vs. true 0.100 rad).
- All 8 optimisation starts converge to the same value ($\log \mathcal{L} \approx 4.098 \times 10^9$), confirming unimodality.

Parameter	Symbol	Default
Detection time	T	3.8 s
GH order per dimension	n_{gh}	4
Total GH nodes per shot per AI	n_{gh}^4	256
QMC samples per GH node	n_{qmc}	2048
Phase grid points	K	128
Optimiser starts	n_{starts}	8
Search domain half-width	A_{max}	0.2 rad
COM prior std (position)	τ_{pos}	10 μm
COM prior std (velocity)	τ_{vel}	10 $\mu\text{m/s}$
Width prior mean (position)	$\bar{\sigma}_{\text{pos}}$	100 μm
Width prior mean (velocity)	$\bar{\sigma}_{\text{vel}}$	100 $\mu\text{m/s}$
Width prior std (position)	$\sigma_{\sigma,\text{pos}}$	10 μm
Width prior std (velocity)	$\sigma_{\sigma,\text{vel}}$	10 $\mu\text{m/s}$

Table 1: Default hyperparameters for the 4D GH moment inference pipeline (`python-scripts/moment_inference.py`).

Stage	Scope	Wall-time (A100)
Load PSMAPs	Once	≈ 25 s
Data reduction + ridge priors	Per shot/AI	< 1 ms
GH grid + ACS precompute	Per shot/AI	≈ 0.56 s
Total precompute (200 shots)	Per run	≈ 1.9 min
$\log \mathcal{L}(\beta)$ single eval	Per eval	≈ 0.5 s
Optimisation ($n_{\text{starts}} = 8$)	Per run	≈ 95 s
Total (200 shots)	Per run	≈ 10 min

Table 2: Wall-times for $N_{\text{shots}} = 200$, $n_{\text{gh}} = 4$, $n_{\text{qmc}} = 2048$, $K = 128$. The precompute stage is dominated by $200 \times 2 \times 256 = 102\,400$ batched PSMAP calls.

8 8D Prior Marginalisation (Moment-Free Baseline)

This section describes a simplified variant of the inference pipeline that *ignores* the observed final-position moments entirely. Rather than collapsing the nuisance integral to a 4D ridge via the $R=0$ Kalman update (Section 7), the cloud parameters η are marginalised directly against the 8D prior $p_0(\eta)$. The result is a strictly *less informative* likelihood that serves as a baseline: comparing it with the moment-conditioned result quantifies how much information the spatial image data contribute beyond the total port counts.

8.1 Reduction to Port Counts Only

For each shot i and each AI z , the only data used are the total port counts $n_{izg} = \sum_b n_{izgb}$ and $n_{ize} = \sum_b n_{izeb}$. No spatial moments are extracted; the full image is reduced to two scalars.

The integrated ACS model (54)–(56) and the port-fraction log-likelihood (57) are used unchanged.

8.2 Marginalisation over the 8D Prior

The log-marginal for shot i at laser phase ϕ and signal β is

$$\log \mathcal{L}_i^{\text{prior}}(\beta) = \log \text{sumexp}_{k=0}^{K-1} \left[\log F_{iz_0}(\phi_k) + \log F_{iz_{100}}(\phi_k + \delta\phi_i(\beta)) \right] - \log K, \quad (69)$$

where

$$\log F_{iz}(\Phi) = \log \text{sumexp}_{j=1}^{n_\eta} \left[\log \tilde{w}_j + \ell_{iz}(\Phi; \eta^{(j)}) \right], \quad \log \tilde{w}_j = -\log n_\eta \quad \forall j, \quad (70)$$

and $\{\eta^{(j)}\}_{j=1}^{n_\eta}$ are fixed samples from the prior $p_0(\eta)$. The uniform weights $\tilde{w}_j = 1/n_\eta$ reflect that the samples are drawn directly from $p_0(\eta)$, so no importance correction is needed.

8.3 Sobol QMC Sampling of the Prior

The prior factorises over the 8 cloud parameters $\eta = (\mu_{x_0}, \mu_{y_0}, \mu_{v_x}, \mu_{v_y}, \sigma_{x_0}, \sigma_{y_0}, \sigma_{v_x}, \sigma_{v_y})$:

$$\begin{aligned} \mu_{x_0}, \mu_{y_0} &\sim \mathcal{N}(0, \tau_{\text{pos}}^2), & \mu_{v_x}, \mu_{v_y} &\sim \mathcal{N}(0, \tau_{\text{vel}}^2), \\ \sigma_{x_0}, \sigma_{y_0} &\sim \mathcal{N}(\bar{\sigma}_{\text{pos}}, \sigma_{\sigma, \text{pos}}^2), & \sigma_{v_x}, \sigma_{v_y} &\sim \mathcal{N}(\bar{\sigma}_{\text{vel}}, \sigma_{\sigma, \text{vel}}^2). \end{aligned} \quad (71)$$

Standard deviations are clipped to a small positive floor after sampling.

Samples are drawn using a scrambled Sobol sequence in $[0, 1]^8$ transformed through the normal quantile function. Sobol sequences achieve a discrepancy of $\mathcal{O}((\log n_\eta)^8/n_\eta)$ in 8D, compared to $\mathcal{O}(1/\sqrt{n_\eta})$ for plain Monte Carlo, yielding significantly better coverage of the prior at the same sample budget. The default $n_\eta = 512$ gives a comparable total evaluation count to the $n_{\text{gh}}^4 = 256$ nodes of the 4D GH rule.

Why not 8D Gauss-Hermite. A product GH rule with n_{gh} nodes per dimension requires n_{gh}^8 evaluations. Even $n_{\text{gh}} = 2$ gives $2^8 = 256$ nodes (only three polynomial degrees per axis — far too coarse for the nonlinear ACS integrand). $n_{\text{gh}} = 4$ would require $4^8 = 65\,536$ nodes, a 256-fold increase over the 4D case, making it computationally infeasible. QMC with $n_\eta = 512$ achieves comparable or better accuracy at the same cost as the 4D moment-conditioned method.

8.4 Computational Cost

Because the prior $p_0(\eta)$ does not depend on any shot-specific data, the integrated ACS need be evaluated at the n_η prior samples *only once* per AI — not once per shot. The precompute is therefore $\mathcal{O}(n_\eta)$ instead of $\mathcal{O}(N_{\text{shots}} \times n_\eta)$, reducing the precompute from ~ 2 min to a few seconds. Only the per-shot port counts $(n_{\text{izg}}, n_{\text{ize}})$ are read in a fast loop.

Stage	Scope	Wall-time (A100)
Load PSMAPs	Once	≈ 25 s
Sample n_η prior points (Sobol)	Once	< 0.1 s
ACS precompute (n_η pts, 2 AIs)	Once	≈ 5 s
Read port counts (200 shots)	Once	< 1 s
$\log \mathcal{L}^{\text{prior}}(\beta)$ eval	Per eval	≈ 0.5 s
Optimisation ($n_{\text{starts}} = 8$)	Per run	≈ 95 s
Total (200 shots)	Per run	≈ 2 min

Table 3: Wall-times for `python-scripts/prior_inference.py`, $N_{\text{shots}} = 200$, $n_\eta = 512$, $K = 128$. The precompute saving over the moment-conditioned method (Section 7) arises because the prior ACS is shot-independent.

8.5 Information Content of Final Moments

Let $\hat{\beta}^{\text{mom}}$ and $\hat{\beta}^{\text{prior}}$ denote the MLEs from the moment-conditioned and prior-only pipelines respectively. Three complementary diagnostics quantify the value of the spatial moment data:

1. **Log-likelihood gain.** The quantity

$$\Delta \log \mathcal{L} = \log \mathcal{L}^{\text{mom}}(\hat{\beta}^{\text{mom}}) - \log \mathcal{L}^{\text{prior}}(\hat{\beta}^{\text{prior}}) \quad (72)$$

is always non-negative (both methods use the same port-count data; the moment-conditioned method uses strictly more information). A large $\Delta \log \mathcal{L}$ means the spatial moments substantially sharpen the nuisance posterior, aiding signal recovery.

2. **Recovery accuracy.** Compare $|\hat{\beta}^{\text{mom}} - \beta_{\text{true}}|$ vs. $|\hat{\beta}^{\text{prior}} - \beta_{\text{true}}|$. Bias or increased variance in the prior-only estimate indicates that moment conditioning improves the identifiability of the signal.
3. **Signal-to-noise ratio.** On repeated datasets (varying shot noise and cloud realisations), the ratio of signal-to-null log-likelihood ratios $\log \mathcal{L}(\hat{\beta}) / \log \mathcal{L}(0, 0)$ from the two pipelines gives the effective SNR gain from moment conditioning.

The prior-only method therefore serves as both a practical fallback (for settings where images are not recorded or not trusted) and as a diagnostic probe of how much spatial information is present in the cloud images.

9 Full Marginal Log-Likelihood

9.1 Rotation Trick for AI $z = 100$

For AI $z = 100$, the effective phase at grid point k in shot i is $\phi_k + \delta\phi_i(\beta)$. Rather than recomputing the ACS for each $(\delta\phi_i, k)$, we apply a rotation to the precomputed ACS:

$$C_{zsb}^{c,\text{eff}}(\beta) = C_{zsb}^c \cos(\delta\phi_i) + C_{zsb}^s \sin(\delta\phi_i), \quad (73)$$

$$C_{zsb}^{s,\text{eff}}(\beta) = -C_{zsb}^c \sin(\delta\phi_i) + C_{zsb}^s \cos(\delta\phi_i). \quad (74)$$

This follows from the angle-addition identity and means that the per-evaluation cost for Z100 is a 2D rotation of the precomputed ACS tensors, not a new PSMAP evaluation.

9.2 Per-Shot Marginal Log-Likelihood

Given precomputed ACS $\{A_{zsb}, C_{zsb}^c, C_{zsb}^s\}$ and scale factor L_{iz} at $\hat{\eta}_{iz}$, the conditional log-likelihood at phase grid point k is:

$$\log \mathcal{L}_i(\beta, \phi_k) = \sum_z \sum_{s \in \{g, e\}} \sum_b \left[n_{izsb} \log \lambda_{izsb}(\phi_k) - \lambda_{izsb}(\phi_k) \right], \quad (75)$$

where for Z0: $\lambda_{igsb}(\phi_k) = L_{i,0} [A_{0sb} + C_{0sb}^c \cos \phi_k + C_{0sb}^s \sin \phi_k]$, and for Z100 the rotated ACS coefficients are used. The marginal over ϕ_i is then (17).

9.3 Full Log-Likelihood

Since shots are independent given β :

$$\log \mathcal{L}(\beta) = \sum_{i=1}^{N_{\text{shots}}} \log \mathcal{L}_i^{\text{marg}}(\beta). \quad (76)$$

10 Implementation

10.1 ACS Pre-Computation (Once per Shot per AI)

The following steps are performed once per shot per AI and the results cached to disk as a compressed NumPy archive (.npz) keyed by `run/site/bins/n_quad/K`:

1. **Cloud nuisance profiling.** Solve (19) with L-BFGS-B (up to 200 iterations, `ftol`= 10^{-10} , `gtol`= 10^{-8} , 20 line-search steps). The objective evaluates the logsumexp over K phase-grid points for both ports jointly at the current η via `SurrogatePixelACS`.
2. **ACS at $\hat{\eta}_{iz}$.** Call `SurrogatePixelACS.pixel_acs( $\hat{\eta}_{iz}$ )` to obtain tensors $\{A_{zsb}, C_{zsb}^c, C_{zsb}^s\}_{s \in \{g,e\}}$ of shape $(N_b,)$ each, on GPU.
3. **Scale factor.** Compute L_{iz} via (16).
4. **Cache.** Store ACS tensors, L_{iz} , $\hat{\eta}_{iz}$, and raw pixel counts (n_{izgsb}, n_{izesb}) to .npz.

10.2 Signal Fit (Per Run)

Given the cached ACS tables for all shots in a run, fit $\beta = (A_s, A_c)$:

1. **Signal phases.** $\delta\phi_i = A_s \sin(2\pi f t_i) + A_c \cos(2\pi f t_i)$ for $i = 1, \dots, N_{\text{shots}}$.
2. **Rotation trick.** For each shot i , rotate the Z100 ACS tensors by $\delta\phi_i$.
3. **Expected counts.** For each phase point ϕ_k , compute $\lambda_{izsb}(\phi_k)$ via (11) as a GPU broadcast: `lam[k, b] = L * max(A[b] + Cc[b]*cos_phi[k] + Cs[b]*sin_phi[k], 1e-300)`.
4. **Conditional logL.** Compute (71) for each (i, k) pair.
5. **Phase marginalisation.** Apply logsumexp over k and subtract $\log K$.
6. **Shot sum and optimisation.** Sum over shots; negate and pass to L-BFGS-B with bounds $A_s \in [0, \pi]$, $A_c \in [-\pi, \pi]$.

10.3 Multi-Start Strategy for β

The marginal log-likelihood $\log \mathcal{L}(A_s, A_c)$ is smooth but can be flat near the null $(0, 0)$ when the signal is weak. We use multi-start L-BFGS-B from amplitude starting values $A_s^{(0)} \in \{0, 0.03, 0.07, 0.10, 0.15\}$ at $A_c^{(0)} = 0$, take the best solution, and report convergence diagnostics.

10.4 Computational Cost

Step	Phase	Typical wall-time
<code>SurrogatePixelACS</code> construction	Once (shared)	~ 12 s
Cloud profiling ($K=512$ logsumexp)	Per shot/AI	~ 0.5 – 2 s
All shots profiling + ACS cache	Per run	~ 1 – 2 min
Signal fit (50 shots, multi-start)	Per run	~ 40 s

Table 4: Typical timings on one GPU (A100/V100 class) with $K = 512$, $N_b = 1024$ (32×32 bins), $n_{\text{quad}} = 20,000$, $N_{\text{shots}} = 50$. Cloud profiling timings dominate on the first pass; subsequent runs load from the .npz cache in milliseconds.

11 Maximum Likelihood Estimation

11.1 Science Parameters

With f fixed, the only free parameters are $\beta = (A_s, A_c)$. The MLE is

$$\hat{\beta} = \arg \max_{\beta} \log \mathcal{L}(\beta), \quad (77)$$

solved by multi-start L-BFGS-B (Section 9). The recovered amplitude and phase are $\hat{A} = \sqrt{\hat{A}_s^2 + \hat{A}_c^2}$ and $\hat{\varphi} = \arctan 2(\hat{A}_c, \hat{A}_s)$.

11.2 True Signal Parameterisation

The simulated signal is generated as $\delta\phi_i = A_{\text{true}} \sin(2\pi f t_i + \varphi_{\text{true}})$. Expanding via angle addition:

$$A_s^{\text{true}} = A_{\text{true}} \cos \varphi_{\text{true}}, \quad A_c^{\text{true}} = A_{\text{true}} \sin \varphi_{\text{true}}, \quad (78)$$

so the ground-truth (A_s, A_c) is read directly from the H5 file attributes `signal_amp` and `signal_phase`. Note that $\delta\phi_i$ is stored only in the Z100 H5 file (it is zero in Z0 by construction).

12 Validation

Running `fit_surrogate_profiled_signal.py` with oracle cloud initialisation (`--init-from-true`) on the default 1 M-atom dataset ($N_{\text{shots}} = 50$, $K = 512$, $n_{\text{quad}} = 20,000$) gives:

- Signal amplitude RMSE ≈ 0.0013 rad (1.3 % of true $A = 0.1$ rad).
- Signal phase RMSE ≈ 0.009 rad.
- Cloud COM RMSE: $\sim 2\text{--}5 \mu\text{m}$ for position, $\sim 1\text{--}4 \mu\text{m/s}$ for velocity.
- Cloud width RMSE: $\sim 0.1\text{--}0.3 \mu\text{m}$.
- L-BFGS-B converges in ~ 5 iterations and ~ 25 function evaluations for both the cloud profiling and the signal fit.